Evidence-Grounded Reliability Evaluation of Vision-Language Models for Chest X-ray Interpretation under Distribution Shift

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Abstract

Vision-language models (VLMs) now write chest X-ray (CXR) reports, and some also return a box for each finding. Their reliability is still measured almost entirely by comparing the generated report with a reference report written at the same hospital. Under distribution shift this comparison is confounded, because a change of hospital changes the reporting style as well as the images, so a drop in a reference-based score cannot be separated from a change in how reports are written. We propose an evaluation that checks every positive finding a VLM claims against independent image evidence: bounding boxes drawn by expert radiologists. We define the Evidence-Support Precision (ESP), the share of claimed findings that are supported by expert evidence in the same image, together with its complement, the unsupported-claim rate, and evidence recall. We use the agreement of a claim's localisation across sampled generations as a confidence signal for abstention. We evaluate open VLMs that localise their own findings (CheXagent-2, Qwen2.5-VL and Lingshu, with MedGemma 1.5 and MAIRA-2 added once access is granted) on six public sites from three countries (USA, China and India) and under five families of image corruption at two severities. We introduce an evidence-ablation test that removes the expert-marked regions from the image, against a control that removes equal areas elsewhere in the lungs, to measure whether a claim depends on its evidence at all. We further test whether conformal claim filters calibrated at one site keep their guarantee at others, and propose a training-free self-grounding filter whose threshold is recalibrated at a new hospital from a few dozen labelled images. Across the six sites, moving from the NIH-derived site to the Chinese TB site lowers evidence

recall for every model (CheXagent-2: 37.6% to 16.3%) far more than it lowers ESP, and models claim fewer findings rather than more unsupported ones. Text-level and box-level support differ by 21–41 points, so a correctly named finding is often not placed on its evidence. The sentence probability ranks supported claims below chance in every model–site setting (AUROC 0.25–0.47), whereas a yes/no probe reaches 0.69–0.93. Qwen2.5-VL's claims are lost as often after an unrelated edit as after the removal of their evidence (evidence-dependence index 4 and **−17** points), against 53 and 58 points for CheXagent-2. Under severe blur, noise or JPEG compression the Qwen-family models assign the image a low quality when asked (P(adequate) down to 0.001), yet no model mentions the degradation in more than 4% of reports. Conformal claim filters calibrated at one site over-cover rather than under-cover at the TB site and discard most claims; recalibrating from 50 local images restores the target coverage while keeping 3.8 times as many claims. Code, prompts and label mappings are available at https://github.com/ailakshya/cxr-vlm-reliability.

Keywords: Vision-language models, Chest X-ray, Radiology report generation, Grounding, Hallucination, Distribution shift, Calibration, Selective prediction, Conformal prediction

1 Introduction

Chest radiography is the most frequently performed imaging examination, and automatic report generation is one of the most studied applications of medical vision-language models (VLMs). In the last two years the field has moved from text-only generators such as LLaVA-Med [1], RadFM [2], LLaVA-Rad [3] and CheXagent [4] to models that attach spatial evidence to what they say. MAIRA-2 produces a list of sentence–box pairs [5], MedGemma 1.5 adds anatomical bounding-box localisation [6], and RadVLM [7] and AnatomiX [8] ground phrases in the image. General-purpose models such as Qwen2.5-VL [9] also return boxes natively.

The way these models are evaluated has not kept pace. Almost every report-quality metric compares the generated report with a reference report: CheXbert F1 [10], RadGraph F1 [11], RadCliQ [12], GREEN [13], RaTEScore [14] and FineRadScore [15]. RadFact, introduced with MAIRA-2, adds a spatial check, but again against reference boxes from the same distribution [5]. Reference-free hallucination detectors exist, but they rely on model-internal signals rather than on the image [16, 17]. FactCheXcker re-examines the image only for quantitative measurements such as tube position [18].

This dependence on reference text becomes a problem when a model is moved to a new hospital, which is the situation that matters for deployment. The CXR literature has shown repeatedly that models degrade across sites [19, 20] and across patient groups [21]. For VLMs the evidence is growing. Weak image artefacts reduced CXR accuracy by 10.46% on average, and models flagged strong artefacts in only 11.5% of cases [22]. Report quality falls on out-of-distribution sets [23, 24], and clinician preference for Flamingo-CXR reports was 56.1% for intensive-care films against 77.7% for outpatient films [25]. But when MAIRA-2 was evaluated on PadChest, its lexical scores

were far below its clinical scores, which the authors attributed partly to a different report structure [5]. The choice of reference report alone changes the scores [26], and Med-Gemini did not report RadGraph F1 on its Indian test set because the metric is tuned to MIMIC-style reports [27]. When the site changes, a reference-based metric cannot tell whether the model got worse or the reports were written differently.

Expert boxes do not have this problem. A box drawn by a radiologist around a pneumothorax is evidence about the image, not about how a report is phrased, and it means the same thing at every hospital. Several public CXR datasets provide such boxes [28–31], but they have been used to score phrase grounding within one distribution rather than to check the faithfulness of generated reports across distributions.

At the same time, reliability benchmarks for medical VLMs have multiplied [32–42], and calibration and abstention methods have been adapted to medical generation [17, 43–47]. The closest prior studies each cover at most two of the three ingredients that matter here: evidence checks, controlled shift, and calibration under that shift. Predictive entropy has been shown to support selective prediction from MIMIC to PadChest, but for binary questions, without grounding [48]. HalluCXR measures hallucination on MIMIC only [40]. MVC-Bench measures calibration under domain shift for contrastive classifiers [41]. RadFact checks grounding but not shift or calibration [5]. Grounded dual-state calibration ties confidence to visual support through counterfactual image substitution in VQA [49].

This paper combines the three. Our contributions are:

1. An image-evidence faithfulness metric for generated CXR reports, Evidence-Support Precision (ESP), with the unsupported-claim rate and evidence recall, computed against expert boxes and independent of reference reports (Section 4.3).
2. A multi-site, multi-severity evaluation of grounding and faithfulness for open CXR VLMs that localise findings natively or on request, including label-level probes on Indian hospital data (Sections 5–6).
3. An evidence-ablation test: the expert-marked regions are replaced by contralateral or blurred tissue, and the change in claims and yes/no probabilities is compared with a matched control that edits equal-sized lung regions without evidence. The difference, the *evidence-dependence index*, separates claims that rest on the marked findings from claims the model would make anyway (Section 4.4).
4. A comparison of evidence-based and uncertainty-based abstention under shift, a test of whether conformal claim filters calibrated at one site keep their coverage at other sites, and a training-free *self-grounding filter* that combines sample agreement, the yes/no probe and box agreement, with its threshold recalibrated at the target site from $m \in \{10, 25, 50\}$ labelled images (Section 4.5).
5. Released code, prompts, and label mappings between model vocabularies and dataset annotations.

We test four hypotheses. **H1**: drops in reference-based metrics across sites overstate or misorder the drops in ESP. **H2**: under image corruption the unsupported-claim rate rises faster than models report degraded image quality. **H3**: localisation agreement across samples is a better abstention signal than token entropy out of distribution.

H4: conformal claim filters calibrated at one site under-cover at others, and a few dozen local labels restore coverage. **H5**: a substantial share of correct claims survives the removal of their evidence about as often as the removal of an unrelated region, so text-level correctness overstates grounding.

2 Related work

Chest X-ray vision-language models.

Contrastive models such as CheXzero [50] and BiomedCLIP [51] align images with report text and give zero-shot classification and similarity heatmaps. Generative models produce free-text reports: LLaVA-Med [1], RadFM [2], LLaVA-Rad [3], CheXagent [4], Lingshu [52] and the Gemini-based Med-Gemini [27]. Grounded report generation was introduced by MAIRA-2 [5] and extended by RadVLM [7], AnatomiX [8] and MedGemma 1.5 [6]. Reinforcement-learning approaches to medical grounding have also appeared [53].

Report evaluation and hallucination.

Report metrics range from label agreement (CheXbert [10]) and entity–relation graphs (RadGraph [11]) to composite and LLM-based scores (RadCliQ [12], GREEN [13], RaTEScore [14], FineRadScore [15]). ReXrank ranks report generators on several data sources [54]. Hallucination benchmarks for medical VLMs include MedVH [34], MedHallMark [35], MedHallBench [36], MedHEval [37] and, for CXR, HalluCXR [40]. ProbMed showed that adversarial probing questions push medical VLMs below random accuracy [33], and CARES evaluates trustworthiness more broadly [32].

Distribution shift in chest radiography.

Site-dependent generalisation [19], shortcut learning [20] and underdiagnosis in underserved groups [21] are well documented for classifiers. For VLMs, robustness has been studied under synthetic corruption [55, 56], realistic artefacts [22], quality degradation [24, 39], misleading context [42], and realistic VQA shifts [38]. Report generation has been evaluated on held-out sources [23, 54], but the score changes have not been attributed to the type of shift or checked against image evidence.

Calibration, abstention and conformal prediction.

Semantic consistency across sampled reports identifies uncertain reports, and abstaining on the most uncertain 20% improves factuality [17]. Conformal Factuality [43] and Conformal Alignment [44] filter generated claims with statistical guarantees, and selective conformal risk control has been extended to shift [45]. In medicine, conformal prediction has been applied to CXR impression drafting [47]. A small number of local labels has been shown to raise certified coverage in medical VQA considerably [46], and predictive entropy transfers as a selective-prediction signal across two CXR sites [48]. MVC-Bench benchmarks calibration of medical contrastive models [41]. None of these uses expert image evidence as the definition of a correct claim. The closest counterfactual test, grounded dual-state calibration, swaps the whole image [49]; removing

Table 1 Datasets. *Role*: E = expert box evidence, R = reference reports, L = image-level labels only. Access: O = open download, F = free registration, P = PhysioNet credentialed, Q = on request. The last three rows were not evaluated in this study (access pending).

Dataset	Country	Images	Annotation	Access	Role
ChestX-Det test [28]	USA (NIH)	553	boxes, 13 classes	O	E
ChestX-Det train (calib.)	USA (NIH)	300	as above	O	E
Shenzhen [57, 58]	China	662	TB; typed lesion outlines	O	E, L
Montgomery [57]	USA	138	TB label	O	L
IU-Xray [59]	USA (Indiana)	500	reports	O	R
Mendeley CXR [60]	India (3 hospitals)	400 of 799	COVID / abnormal / normal	O	L
NITRD TB A/B [61]	India (New Delhi)	244	TB / non-TB, 2 scanners	O	L
MIMIC-CXR + MS-CXR [5]	USA (Boston)	–	reports, phrase boxes	P	R, E
VinDr-CXR [30]	Vietnam	–	boxes, 22 findings	P	E
PadChest-GR [31, 62]	Spain	–	grounded reports	Q	R, E

only the expert-marked region, with an equal-area control, asks a narrower question: whether the claim depends on the specific evidence.

3 Datasets and shift axes

Table 1 lists the data. Expert boxes provide the evidence, and report-bearing sets let us compare ESP with reference-based metrics. Label-only sets serve as classification-style probes.

Site.

Each source is treated as a site. The two evidence sites differ in country, scanner and annotation protocol. In ChestX-Det the boxes of 13 classes are mapped to the shared space $\mathcal{F}$; calcification and fibrosis have no counterpart and support only the generic *opacity* claim, and emphysema supports no claim. In Shenzhen, two radiologists outlined every lung lesion of the TB-positive films and named its type [58]; infiltrates map to consolidation, nodules and cavities to nodule/mass, and effusion and apical, interlobar or other pleural thickening to their counterparts. Normal Shenzhen films have no lesions, so every finding is absent there; on TB films the heart and bones were not reviewed, so cardiomegaly and fracture claims are excluded as unverifiable. Indian data have image-level labels only and are evaluated as a normal/abnormal or TB probe (accuracy, calibration and selective risk), not with ESP. MIMIC-CXR, on which most models were trained, would serve as an in-distribution anchor once access is granted.

Table 2 Models. Precision is the numeric format used for inference.

Model	Parameters	Box output	Precision	Licence
CheXagent-2 [4]	3B	native, 0–100 grid	bf16	MIT
Qwen2.5-VL [9]	7B	native, JSON pixels	FP8	Apache-2.0
Lingshu [52]	7B	prompted, JSON pixels	FP8	MIT
MedGemma 1.5 [6]	4B	native	bf16	HAI-DEF terms (pending)
MAIRA-2 [5]	7B	native, grounded report	8-bit	research only (pending)

Image quality.

Following MedMNIST-C [55], we apply five corruption families at a mild and a severe level to 100 images from each evidence site: additive Gaussian noise ($\sigma = 0.04, 0.16$ on a $[0, 1]$ scale), Gaussian blur ($\sigma = 1, 4$ pixels per 512 pixels of image size), JPEG compression (quality 30, 5), contrast reduction (factor 0.6, 0.25 around mid-grey) and rotation ($5°, 20°$). Expert boxes are rotated with the image.

Acquisition and demographics.

View position and sex and age groups need metadata that only the credentialed sites carry, so they are left for future work. Within the open sites, NITRD Datasets A (152 images, 8-bit JPEG) and B (92 images, 14-bit DICOM) were acquired on two different scanners.

4 Method

4.1 Models

We evaluate open models that fit on one 16 GB GPU (Table 2). All of them return boxes as text, so a claim can be localised by asking the model to locate the claimed finding. CheXagent-2 was trained to emit boxes on a 0–100 grid and sometimes grounds its report sentences directly; Qwen2.5-VL emits JSON boxes in pixel coordinates; Lingshu shares the Qwen2.5-VL architecture but was not evaluated on medical grounding by its authors, so its boxes test whether the format alone transfers. MedGemma 1.5 and MAIRA-2 are gated and are added when access is granted. Post-hoc saliency for text-only models is left for future work. CheXagent-2's instruction data include NLM-TB (Shenzhen and Montgomery), OpenI (IU-Xray) and ChestX-ray14, from which the ChestX-Det images are drawn [4]; for this model only the two Indian sites are unseen hospitals, and results on the other sites are read with that in mind. The training data of the Qwen-family models are not published.

4.2 Claim extraction

For image x, model m generates a findings section r. A CheXbert labeller [10] run locally extracts the set of positive findings. Each finding f is mapped to a shared label space $\mathcal{F}$ (atelectasis, cardiomegaly, consolidation, effusion, pneumothorax, nodule/mass, fracture, pleural thickening, opacity), using hand-built mappings to each dataset's label set (Appendix A). If the model localises a finding, we attach its box $\hat{b}$.

Grounding tags in CheXagent-2 output are removed before labelling. This gives the claim set

$$C_m(x) = \{(f_j, \hat{b}_j)\}_{j=1}^{|C_m(x)|}, \qquad f_j \in \mathcal{F},\ \hat{b}_j \in \mathcal{B} \cup \{\varnothing\}. \tag{1}$$

We check 200 extracted claims by hand and report the extraction error rate. Uncertain mentions ("possible effusion") are recorded but not counted as claims.

4.3 Evidence-support metrics

Let $E(x) = \{(f, B_f)\}$ be the expert evidence for image x, where B_f is the set of expert boxes for finding f. We assume that, within $\mathcal{F}$, the annotation is exhaustive: a finding without a box is absent.

Definition 1 (Supported claim) *A claim $(f_j, \hat{b}_j)$ is supported by $E(x)$ if f_j has at least one expert box and, when $\hat{b}_j \neq \varnothing$, $\max_{B \in B_{f_j}} \mathrm{IoU}(\hat{b}_j, B) \geq \tau$. The pointing variant replaces the IoU test by requiring the centre of a predicted box to lie inside an expert box.*

Over a set of images X,

$$\mathrm{ESP}_m = \frac{\sum_{x \in X} \left|\{c \in C_m(x) : c \text{ supported}\}\right|}{\sum_{x \in X} |C_m(x)|}, \qquad \mathrm{UCR}_m = 1 - \mathrm{ESP}_m, \tag{2}$$

$$\mathrm{ER}_m = \frac{\sum_{x \in X} \left|\{f : B_f \neq \varnothing,\ f \text{ claimed and supported}\}\right|}{\sum_{x \in X} \left|\{f : B_f \neq \varnothing\}\right|}. \tag{3}$$

We report $\tau \in \{0.3, 0.5\}$, and a text-only variant ($\tau = 0$) that ignores location. We also report standard grounding scores (mIoU, accuracy at IoU 0.3 and 0.5, pointing game) and, where reference reports exist, CheXbert F1, RadGraph F1 and GREEN. On corrupted images we record the *artefact-awareness rate*: the share of reports that mention degraded image quality.

4.4 Evidence ablation

For an image with expert evidence, the *evidence-removed* version replaces every expert box by the mirrored region of the opposite hemithorax, or by a heavily blurred copy of the image (Gaussian, $\sigma = 0.08$ of the image size) when the mirrored region holds a lesion too. The *control* version applies the same fill to boxes of the same sizes placed at random inside the lung zones (left and right of the mediastinum, 10–82% of the image height) so that they touch no expert box. Only images for which every control box can be placed are used, so both versions edit the same number of boxes of the same sizes. For every finding f with evidence that the model claims on the clean image, we record whether the claim survives each edit, and for every evidenced finding the drop in the yes/no probability Δp_f. The evidence-dependence index is

$$\mathrm{EDI} = \Pr[\text{claim lost} \mid \text{evidence removed}] - \Pr[\text{claim lost} \mid \text{control}], \tag{4}$$

and analogously for Δp_f. An EDI near zero means the model's claims and confidence are insensitive to the specific evidence, even when the claims are correct.

4.5 Confidence signals and abstention

For every claim we compute four confidence scores: (i) the probability of the claim sentence, the exponential of its mean token log-probability in the greedy report; (ii) semantic consistency, the share of K sampled reports that contain the same finding [17]; (iii) a yes/no probe, the probability the model gives to "yes" against "no" when asked whether the image shows the finding; and (iv) *evidence consistency*,

$$s_{\mathrm{ev}}(f) = \frac{1}{K}\sum_{k=1}^{K} \mathbb{1}[f \in C^{(k)}] \cdot \overline{\mathrm{IoU}}\big(\hat{b}_f^{(k)}, \hat{b}_f^{(k')}\big), \tag{5}$$

computed over one greedy and K' sampled localisation requests for f: it multiplies the share of requests that return a box by the mean pairwise IoU of the returned boxes. The *self-grounding score* is the mean of (ii), (iii) and (iv), the three signals that condition on the image rather than on the generated text alone.

With a claim counted correct when it is supported at the text level, we report expected calibration error (ECE), Brier score, area under the risk–coverage curve (AURC), and selective risk at 80% and 90% coverage, for each site and severity.

Conformal transfer.

Following Conformal Factuality [43], for each calibration report we take the smallest threshold on the confidence score above which every retained claim is supported. The $(1-\alpha)$ quantile $\hat{q}$ of these thresholds, computed on held-out ChestX-Det training images, guarantees under exchangeability that a filtered report contains no unsupported claim with probability at least $1-\alpha$. We compute $\hat{q}$ there and measure the realised coverage and the retained-claim fraction at every other site and severity, for $\alpha \in \{0.05, 0.1, 0.2\}$. For the *site-adaptive* variant, $\hat{q}$ is recomputed from m randomly drawn labelled images of the target site and evaluated on the remaining images of that site (200 random draws per m).

5 Experimental setup

All sites use every image except Mendeley (400 of 799, drawn with a fixed seed) and IU-Xray (500 studies with a findings section; IU-Xray has no view metadata, so the frontal image of each study is the one BiomedCLIP [51] rates most likely to be frontal in a zero-shot test, and studies without a likely frontal image are skipped). Images are resized so that the longer side is at most 1024 pixels. For each image a model writes one greedy report and, on evidence sites, $K = 5$ sampled reports (temperature 0.7, top-p 0.95). Each finding that is claimed or has expert evidence is localised once greedily and $K' = 3$ times by sampling. The model also answers a yes/no question for each finding in $\mathcal{F}$ and three further questions (is the image normal, does it show tuberculosis, is the image quality adequate). On label-only sites no reports are sampled

Algorithm 1 Evaluation protocol for one model m (corruption runs; ablation runs replace c by the evidence or control edit)

for each site s and corruption (family, severity) c, including none **do**
  for each image x in the sample of site s **do**
    $\tilde{x} \leftarrow c(x)$; transform expert boxes if c is geometric
    generate one greedy report and K sampled reports from $m(\tilde{x})$
    extract claims $C_m(\tilde{x})$; ask for boxes of each claimed or expert-evidenced finding
    compute the four confidence scores for each claim
  end for
  compute ESP, UCR, ER, grounding, reference-based scores and calibration on (s, c)
end for
calibrate a conformal threshold on the calibration site; measure coverage on all (s, c)

and no boxes are requested. The corruption grid uses 100 images per evidence site with $K = 3$ and $K' = 2$; the evidence-ablation runs use every matched image with $K = 3$ and no localisation requests. Sampling settings stored in a checkpoint's generation configuration are ignored, so only the values above apply; Lingshu's configuration sets top-$k = 1$, which would make its samples nearly identical to the greedy report. Prompts are listed in Appendix B. All runs use one NVIDIA RTX 5070 Ti (16 GB). CheXagent-2 runs in bf16 with Transformers 4.40, as its code requires. Qwen2.5-VL and Lingshu are served by vLLM 0.32 with FP8 weights; eight images are processed per batch with batch-invariant kernels, so an image's greedy report, claims and yes/no probabilities do not depend on the other images in the batch and every image has a fixed sampling seed: on 20 images run at one and at eight images per batch, all greedy reports, yes/no probabilities and box answers were identical, and the sampled reports were identical for 17 of the 20 images. This serving setup is about 15 times faster than 4-bit inference with Transformers (2.4 vs. 35.5 s per image with $K = 5$, $K' = 3$), which made the full protocol feasible on one GPU. Differences between models and between sites are tested with paired bootstrap confidence intervals over images (1,000 resamples). Closed commercial models are not evaluated on credentialed data, in keeping with the PhysioNet data use agreement.

6 Results

6.1 Evidence support across sites (H1)

Moving from ChestX-Det to Shenzhen lowers evidence recall for every model: from 37.6% to 16.3% for CheXagent-2, from 6.4% to 1.8% for Qwen2.5-VL and from 14.0% to 7.1% for Lingshu. Precision changes much less. CheXagent-2's ESP falls from 64.9% [62–68] to 55.5% [50–60], Lingshu's from 66.1% [61–71] to 60.2% [52–67], and Qwen2.5-VL's rises from 57.7% [50–65] to 66.7% [52–82] on only 30 verifiable claims. At the new

Table 3 Evidence support on clean images: Evidence-Support Precision at the text level (ESP, with 95% bootstrap interval) and with the model's own box overlapping an expert box (ESP@0.3), evidence recall (ER), and oracle grounding mIoU. Values in %.

	ChestX-Det (USA)				Shenzhen (China)			
Model	ESP	ESP@0.3	ER	mIoU	ESP	ESP@0.3	ER	mIoU
CheXagent-2	64.9 [62–68]	37.8	37.6	0.26	55.5 [50–60]	34.8	16.3	0.24
Qwen2.5-VL	57.7 [50–65]	32.7	6.4	0.22	66.7 [52–82]	40.0	1.8	0.09
Lingshu	66.1 [61–71]	25.5	14.0	0.16	60.2 [52–67]	19.5	7.1	0.09

site the models claim fewer findings (1.60, 0.41 and 0.64 claims per image on ChestX-Det; 0.51, 0.17 and 0.20 on Shenzhen) rather than more unsupported ones, so the shift shows up as missed evidence, which a precision-only hallucination metric would not register. Localisation follows the same split: CheXagent-2, trained to ground, keeps its oracle mIoU (0.26 and 0.24), while the two general models halve theirs (0.22 to 0.09 and 0.16 to 0.09). Text-level and box-level support differ by 21–41 points for every model and site, and by about 40 points for Lingshu, whose box format is only prompted, so a claim that names the right finding is often not placed on it.

The only reference-based score available on the open data, CheXbert micro-F1 against IU-Xray reports, orders the models differently: CheXagent-2 0.39, Lingshu 0.12, Qwen2.5-VL 0.10. The four-fold gap reflects mainly how many findings each model reports (0.61 claims per IU-Xray image for CheXagent-2 against 0.09 and 0.25) and, for CheXagent-2, that IU-Xray is in its training data. ESP shows that the claims the general models do make are about as often supported as CheXagent-2's. A reference-based score therefore mixes recall, precision and training overlap, while ESP and evidence recall separate them (H1). A direct test of H1 across sites requires a site with both reference reports and expert boxes (MS-CXR or PadChest-GR), which we could not access.

6.2 Image corruption and artefact awareness (H2)

Image corruption does not reliably raise the unsupported-claim rate on ChestX-Det (Fig. 1, top). CheXagent-2 stays between 30% and 39% against 35.1% on clean images; the Qwen-family rates move between 19% and 63% without a consistent direction, but each of these points rests on 9–79 claims from 100 images and is noisy. On Shenzhen, CheXagent-2's rate rises under all ten corruptions, from 44.5% to 48–60%, most under severe noise (60.0%) and severe JPEG compression (58.1%). The clearer effect is on what the models say about the image. When asked directly, the Qwen-family models recognise severe degradation: their probability that the image quality is adequate falls from 0.72–0.74 on clean images to 0.001–0.02 under severe blur, 0.08–0.23 under severe noise and 0.20–0.26 under severe JPEG compression (Fig. 1, bottom). CheXagent-2 barely notices (0.69 clean, 0.57–0.62 under the same corruptions). Yet no model mentions degraded quality in more than 4% of its reports under any corruption, and usually in none. Lingshu also writes fewer findings under severe noise and blur (0.09

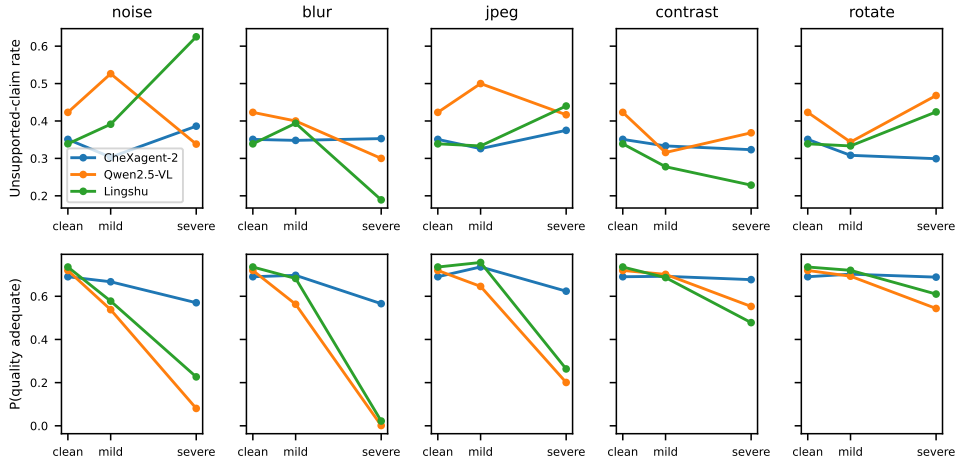


Fig. 1 ChestX-Det under five corruptions at a mild and a severe level: unsupported-claim rate (top) and the model's probability that the image quality is adequate for diagnosis (bottom).

Table 4 Separating supported from unsupported claims: AUROC / AURC of each confidence signal on clean images (higher AUROC and lower AURC are better).

Model	Site	Sent. prob.	Agreement	Yes/no	Box agr.	Self-ground.
CheXagent-2	ChestX-Det	0.46 / 0.38	0.69 / 0.23	0.69 / 0.22	0.62 / 0.27	0.71 / 0.21
CheXagent-2	Shenzhen	0.47 / 0.48	0.54 / 0.40	0.79 / 0.28	0.70 / 0.35	0.76 / 0.29
Qwen2.5-VL	ChestX-Det	0.25 / 0.63	0.78 / 0.21	0.85 / 0.18	0.50 / 0.43	0.80 / 0.22
Qwen2.5-VL	Shenzhen	0.29 / 0.50	0.92 / 0.09	0.93 / 0.09	0.54 / 0.38	0.92 / 0.09
Lingshu	ChestX-Det	0.44 / 0.37	0.76 / 0.17	0.71 / 0.20	0.59 / 0.27	0.76 / 0.18
Lingshu	Shenzhen	0.34 / 0.54	0.74 / 0.23	0.77 / 0.23	0.51 / 0.37	0.74 / 0.21

and 0.37 claims per image against 0.64), without saying why. A reader of the report cannot tell that the model judged the image inadequate (H2).

6.3 Abstention under shift (H3)

The sentence probability, the signal most often used to flag hallucinated sentences, separates supported from unsupported claims worst in all six model–site settings (AUROC 0.25–0.47). It is below chance throughout and far below for Qwen2.5-VL (0.25 on ChestX-Det, 0.29 on Shenzhen): the sentences a model writes most confidently are more often unsupported. The yes/no probe is the best single signal in four of six settings (0.77–0.93), ties with sample agreement for CheXagent-2 on ChestX-Det (0.69), and is within 0.05 of the best in the remaining one. Its advantage holds under the site shift and under corruption: the probe ranks claims better than the sentence probability in all 47 corrupted model–site–corruption settings that contain both supported and unsupported claims. Box agreement beats the sentence probability in every

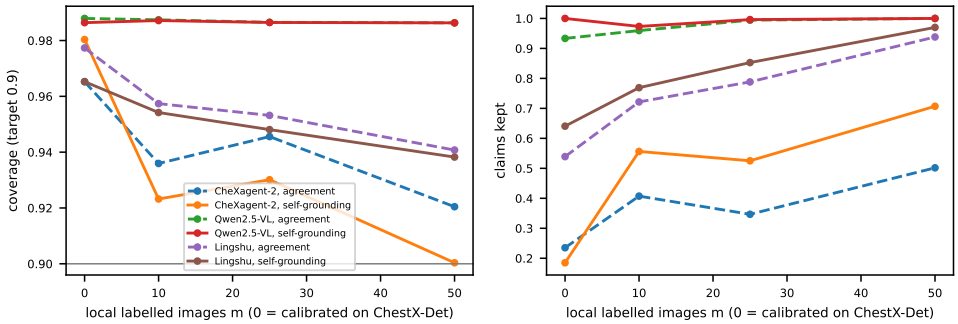


Fig. 2 Claim filter on Shenzhen at target coverage 0.9 ($\alpha = 0.1$): realised coverage (left) and share of claims kept (right) when the threshold is calibrated on ChestX-Det ($m = 0$) or on m labelled Shenzhen images.

clean setting, but it is informative only for the model trained to ground (CheXagent-2, 0.62–0.70; 0.50–0.59, close to chance, for the others). H3 therefore holds against the sentence probability but not against the probe. The self-grounding score, which averages the probe with sample and box agreement, is the best or joint-best signal in two settings (CheXagent-2 on ChestX-Det, 0.71; Lingshu on ChestX-Det, 0.76) and within 0.03 of the probe in three more. Including box agreement costs little even for the general models, because it is one of three averaged terms.

6.4 Conformal transfer (H4) and the site-adaptive filter

On the calibration site's own test images the conformal filters reach their target: at $\alpha = 0.1$ the realised coverage on ChestX-Det is 0.89–0.96 for every model and score. At Shenzhen they do not under-cover, as H4 predicted, but over-cover: 0.96–1.00 at $\alpha = 0.1$, and the self-grounding filter is above target at $\alpha = 0.05$ and 0.2 as well. The guarantee holds but at a cost. CheXagent-2's self-grounding filter keeps only 18.5% of its claims at Shenzhen and its yes/no filter 20.1%, and Lingshu's self-grounding filter keeps 64.1%. Part of the over-coverage is mechanical: Shenzhen reports contain fewer claims (0.51 against 1.60 per image for CheXagent-2), and a report without a claim is covered trivially. Recalibrating the threshold on m labelled Shenzhen images fixes the cost without losing the guarantee (Fig. 2). With $m = 50$, coverage stays at or above 0.90 for every model with both the agreement and the self-grounding score, and CheXagent-2's self-grounding filter keeps 70.7% of its claims instead of 18.5%, and Lingshu's 97.0% instead of 64.1%. Qwen2.5-VL makes so few unsupported claims at Shenzhen that its filter already keeps nearly all of them. Transferred filters are therefore safe but wasteful, and a few dozen local labels recover most of the discarded claims (H4, with the direction of the transfer error reversed).

Table 5 Evidence ablation. Claims made on the clean image that are lost when the expert-marked regions are removed (Evidence) or when equal areas elsewhere in the lungs are removed (Control); their difference (EDI); and the evidence-dependence index of the yes/no probability with its 95% bootstrap interval.

Model	Site	Claims	Lost (evid.)	Lost (ctrl.)	EDI	EDI of p
CheXagent-2	ChestX-Det	275	75.6	22.5	53.1	0.187 [0.170, 0.207]
CheXagent-2	Shenzhen	114	73.7	15.8	57.9	0.212 [0.194, 0.230]
Qwen2.5-VL	ChestX-Det	25	72.0	68.0	4.0	0.050 [0.035, 0.066]
Qwen2.5-VL	Shenzhen	6	50.0	66.7	-16.7	0.009 [-0.005, 0.023]
Lingshu	ChestX-Det	80	95.0	56.2	38.7	0.193 [0.170, 0.215]
Lingshu	Shenzhen	33	100.0	90.9	9.1	0.181 [0.160, 0.206]

6.5 Evidence dependence (H5)

Removing the expert-marked regions removes most of CheXagent-2's claims that rest on them: 75.6% of its evidenced claims on ChestX-Det and 73.7% on Shenzhen, against 22.5% and 15.8% when regions of the same sizes are edited elsewhere in the lungs (Table 5). Its evidence-dependence index is 53 and 58 points on claims and 0.187 [0.170, 0.207] and 0.212 [0.194, 0.230] on the yes/no probability. Qwen2.5-VL behaves differently. It loses its evidenced claims about as often under the control edit as under evidence removal (68% vs. 72% on ChestX-Det, 67% vs. 50% on Shenzhen; EDI 4 and −17 points, on 25 and 6 claims), and its yes/no probability barely depends on the evidence (EDI 0.050 [0.035, 0.066] and 0.009 [−0.005, 0.023]). Lingshu's probability depends on the evidence as much as CheXagent-2's (0.193 and 0.181), but its generated claims are unstable under any edit: it drops 95% and 100% of them when the evidence is removed and 56% and 91% under the control. Two further findings hold across models. About a quarter of CheXagent-2's evidenced claims (24.4% and 26.3%) survive the removal of all their evidence, so they are made without it. And 16–91% of claims disappear after an edit that touches no evidence. Text-level correctness counts both groups as correct (H5).

6.6 Label-level probes, including Indian hospital data

CheXagent-2 detects TB almost perfectly on Montgomery and Shenzhen (AUROC 1.00 and 0.99), but both sites are in its instruction data (Section 4.1), so these figures may partly measure recall of training data. On the unseen Indian NITRD site it still reaches 0.95 (accuracy 86.1%), with a higher calibration error (ECE 0.11 against 0.03–0.04). So most of its TB detection transfers to an unseen hospital, with a small loss. On the unseen Mendeley site, where the task is abnormal versus normal and most abnormal films are COVID-19 cases, its probability is barely informative (AUROC 0.58, accuracy 51.7%, ECE 0.36). Together the two Indian sites show that the large drop comes from the change of task, not from the change of hospital. Lingshu, whose training data are not published, detects TB well at all three TB sites (0.94, 0.86 and 0.95) and is well calibrated (ECE 0.03–0.09). Qwen2.5-VL is moderate everywhere (0.66–0.71) and poorly calibrated on TB (ECE 0.38–0.50). Its 89.2% accuracy on

Table 6 Label-level probes from the yes/no questions: AUROC and accuracy of P(TB) or P(abnormal) = 1−P(normal), calibration error of the answer confidence, and accuracy of the claims in the generated report (abnormal if any finding is claimed).

Model	Site	Label	AUROC	Acc.	ECE	Report acc.
CheXagent-2	Montgomery (USA)	TB	1.00	100.0	0.04	85.5
CheXagent-2	Shenzhen (China)	TB	0.99	97.1	0.03	67.8
CheXagent-2	Mendeley (India)	Abnormal	0.58	51.7	0.36	53.2
CheXagent-2	NITRD (India)	TB	0.95	86.1	0.11	85.7
Qwen2.5-VL	Montgomery (USA)	TB	0.67	58.0	0.42	48.6
Qwen2.5-VL	Shenzhen (China)	TB	0.66	49.2	0.50	52.0
Qwen2.5-VL	Mendeley (India)	Abnormal	0.71	89.2	0.08	42.0
Qwen2.5-VL	NITRD (India)	TB	0.71	59.4	0.38	60.2
Lingshu	Montgomery (USA)	TB	0.94	88.4	0.05	73.9
Lingshu	Shenzhen (China)	TB	0.86	77.8	0.03	58.9
Lingshu	Mendeley (India)	Abnormal	0.75	81.0	0.09	46.5
Lingshu	NITRD (India)	TB	0.95	87.3	0.09	79.1

Mendeley equals the share of abnormal films in the sample (89.3%): it calls almost every film abnormal. On NITRD, the 14-bit DICOM scanner (Dataset B, 92 images) gives a higher TB AUROC than the 8-bit JPEG scanner (Dataset A, 152 images) for all three models (CheXagent-2 0.98 vs. 0.92, Qwen2.5-VL 0.77 vs. 0.66, Lingshu 0.99 vs. 0.93). The subsets are small and differ in TB prevalence (27% vs. 49%), so this is an observation, not a scanner effect we can isolate.

7 Discussion

Correct is not the same as grounded.

Three measurements separate a claim that names the right finding from a claim that rests on the evidence for it. Between 21 and 41 points separate text-level from box-level support, so models often name a finding they do not place on the marked region. A quarter of CheXagent-2's evidenced claims survive the removal of all their evidence, and Qwen2.5-VL's claims are lost as often after an unrelated edit as after evidence removal, so they are made without it. And 16–91% of claims vanish after an edit that touches no evidence, so many claims are fragile rather than wrong. Reference-based metrics, and text-level ESP itself, count all three groups as correct. The evidence ablation with a matched control is cheap (two extra reports per image, no localisation) and could be added to any grounded benchmark.

Which confidence to trust.

The token probability of the generated sentence, the most common hallucination signal, is below chance at separating supported from unsupported claims for every model, and strongly inverted for Qwen2.5-VL. Asking the model a direct yes/no question about the claimed finding is the best or near-best signal in every setting, on clean and corrupted images, and costs one forward pass per finding. Box agreement helps only

for a model trained to ground. A deployment that cannot tune the signal per model should use the yes/no probe, never the sentence probability.

Seen and unseen hospitals.

The public TB sites are in CheXagent-2's instruction data. Its TB detection drops only from 0.99–1.00 to 0.95 at the unseen Indian TB site, but to 0.58 at the unseen Indian site that asks a different question (abnormal versus normal). Without the second Indian TB site, the drop on Mendeley would have been read as a failure to transfer to Indian hospitals. A reliability study therefore has to check each model's training sources, keep at least one unseen site, and hold the task fixed across sites so that a change of hospital is not confounded with a change of question.

Shift as silence.

Under both site shift and image corruption, the models tend to say less rather than hallucinate more: fewer claims, lower evidence recall, and, for the Qwen-family models, a confident judgement that the image is inadequate that never reaches the report. Precision-style hallucination metrics, ESP included, miss this, which is why we report evidence recall and the quality probe alongside them. The same effect makes transferred conformal filters conservative: they keep their guarantee but discard most claims until recalibrated on a few dozen local images.

Limitations.

ESP assumes the expert annotation is exhaustive within the shared label space; findings missed by annotators count against the model. The open grounding data come mostly from one source (NIH); the PhysioNet and PadChest-GR sites are needed for a multi-country evidence comparison. Indian data have labels only, so faithfulness and grounding cannot yet be measured on Indian patients. Claim extraction by an automatic labeller introduces errors, which we quantify by manual review. The 7B models run with FP8 weights, which changes their outputs relative to 4-bit or bf16 inference. Some cells rest on few claims: the general models make 30 verifiable claims or fewer at Shenzhen in some conditions, and the corruption grid uses 100 images per setting, so those rates carry wide intervals. MedGemma 1.5 and MAIRA-2 could not be evaluated because access was not granted in time.

8 Conclusion

We proposed evaluating chest X-ray VLMs against independent expert evidence rather than against reference reports, and applied it to three open models on six public sites from three countries and under ten image corruptions. Under site shift and corruption the models mainly fail by omission: evidence recall drops by half or more while claim precision changes little, and the models detect severe image degradation when asked but never report it. Text-level correctness overstates grounding: 21–41 points separate naming a finding from placing it, and the evidence ablation shows that one model's claims do not depend on the evidence at all. For abstention, the sentence probability is

worse than chance, while a direct yes/no probe works across models, sites and corruptions. Conformal claim filters keep their guarantee across sites but become wasteful, and 50 local labels recover most of the discarded claims. The evidence ablation and the recall-aware metrics are cheap to add to any grounded benchmark. The next step is to apply them to credentialed sites with both reference reports and expert boxes, and to models with a native grounding objective.

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Declarations

- **Funding:** None / to be stated.
- **Conflict of interest:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** This study uses only publicly available, de-identified datasets under their data use agreements. No new data were collected from patients.
- **Consent for publication:** Not applicable.
- **Data availability:** All datasets are available from their original providers under the access terms given in Table 1.
- **Code availability:** Code, prompts, label mappings, raw model outputs and run logs are available at https://github.com/ailakshya/cxr-vlm-reliability.
- **Author contribution:** To be stated.

Appendix A Label mappings

Model reports are labelled by CheXbert into 14 classes, which map to $\mathcal{F}$ as follows: Atelectasis → atelectasis; Cardiomegaly → cardiomegaly; Consolidation and Pneumonia → consolidation; Pleural Effusion → effusion; Pneumothorax → pneumothorax; Lung Lesion → nodule/mass; Fracture → fracture; Pleural Other → pleural thickening; Lung Opacity → opacity. Edema and Enlarged Cardiomediastinum claims are recorded but excluded from ESP, since no evidence site has boxes for them; Support Devices and No Finding are not claims.

ChestX-Det. Atelectasis, Cardiomegaly, Consolidation, Effusion, Pneumothorax, Fracture and Pleural Thickening map to their counterparts; Nodule, Mass and Diffuse Nodule to nodule/mass. A generic opacity claim is supported by any box of Atelectasis, Consolidation, Nodule, Mass, Diffuse Nodule, Fibrosis or Calcification. Emphysema supports no claim.

Shenzhen. Lesion types map as infiltrate, consolidation and pneumonia → consolidation; nodule, mass and cavity → nodule/mass; pleural effusion → effusion; pleural and interlobar thickening → pleural thickening; atelectasis and collapse → atelectasis; pneumothorax → pneumothorax. All parenchymal lesions also support opacity; calcifications, calcified lymph nodes, linear densities and miliary patterns support opacity only; adenopathy, retraction and unspecified lesions are ignored.

Probe questions. Each finding in $\mathcal{F}$ is asked about by name ("pleural effusion", "lung nodule or mass", "rib fracture", "lung opacity"; the others by their label), plus normality, tuberculosis and image quality.

Appendix B Prompts

Report. Qwen2.5-VL and Lingshu: "You are an expert radiologist. Write the findings section of a radiology report for this chest X-ray. Describe only what is visible in the image." CheXagent-2: "Given the indication: "Routine chest radiograph.", write a structured findings section for the CXR."

Localisation. Qwen2.5-VL and Lingshu: "Locate every region showing {finding} in this chest X-ray. Output the bounding box coordinates in JSON format." CheXagent-2: "Locate areas in the chest X-ray where {finding} are present, using bounding box coordinates."

Yes/no probes. "Does this chest X-ray show {finding}?", "Is this chest X-ray normal?", "Does this chest X-ray show signs of tuberculosis?" and "Is the image quality of this chest X-ray adequate for diagnostic interpretation?", each followed by "Answer with yes or no."

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